

Summary

Skilled and results-driven Network Performance and Optimization (NPO) Engineer with 7.4 years of experience working with Vodafone UK in ensuring optimal network performance and service reliability. Specializing in the analysis and troubleshooting of Key Performance Indicators (KPIs) such as CSSR, DCR, Inter/Intra HO failure, and S1 SRR. Proficient in leveraging Linux-based systems for network optimization, automation, and monitoring. Adept at identifying and resolving performance bottlenecks using cutting-edge tools and technologies. Excellent analytical and problem-solving skills with a strong ability to collaborate with cross-functional teams to drive improvements in mobile network services.

Skills and Competencies

Languages/Scripting: Shell/Bash, Python

Data Analysis & Webserver: MySQL, Tableau, Apache, Apache Tomcat

Monitoring/Logging and NOC Tools: BMC Remedy, Netcool, Mycom, Unison, ENM (Ericsson OSS), U2020 (Huawei OSS), NetAct (Nokia OSS), Wireshark, Putty

Virtualization Software: VM Ware, Virtual Box, Citrix

OS: Windows, Unix/Linux [CentOS-7, RHEL-6, 7, 8]

Networking: TCP/IP, OSI, HTTP, SMTP, QoS

Experience

Sr. NPO Engineer in Vodafone India Services Pvt Ltd - Bangalore, India

Oct 2022 – Present

Role – Deputy Manager

- Perform comprehensive analysis and troubleshooting of network performance KPIs including CSSR, DCR, Inter/Intra HO failure, and S1 SRR to maintain service quality for Vodafone UK's mobile network.
- Conduct root cause analysis (RCA) on network degradation incidents using historical data, performance statistics, and logs collected from various network elements.
- Leverage Linux environments for running custom scripts (Bash, Python) that automate the collection and analysis of network performance data to improve the speed of incident detection and resolution.
- Monitor system logs and network health via custom Bash scripts running on Linux servers, ensuring accurate tracking of KPIs and minimizing downtime.
- Optimize and automate performance reporting systems by creating scheduled cron jobs to gather and process network statistics across Linux based RAN servers, providing real-time updates to management teams.
- Troubleshoot network failures and performance degradation using CLI to debug issues, such as failed handovers or high call drop rates.
- Collaborate closely with cross-functional teams (Deployment, Operations, and Change Management) to develop and implement optimization strategies aimed at improving network reliability and performance.
- Work with the network engineering team to resolve issues related to RAN, S1 interface, and mobility protocols using Linux utilities such as `netstat`, `tcpdump`, and `iftop` to analyze network traffic using Wireshark.
- Developed automated script to take backup of useful files and create email notification in case of backup failure.
- Developed script to notify engineers if disk utilization is above the threshold.

RAN Engineer in Ericsson Global India Pvt Ltd - Noida, India

Dec 2021 – Oct 2022

Project: Telstra Australia (RAN)

- Conduct Network Surveillance, take ownership of fault, manage, and resolve fault with minimal outage.
- Fault management for Node-B, eNode-B & NR Sites and Managed elements like RU, RRU, TMA, SIU related faults.

- Worked on baseband like Ericsson BB 6630, 6640, 5216, 5212, DUW and DUS.
- Collaborated with senior engineers to optimize network performance by analyzing system logs and using Linux-based utilities like `dmesg` and `journalctl` for deeper insight into service disruptions.
- Worked with NMS and performance monitoring tools, collecting data from various RAN elements and generating weekly and monthly reports on network performance.
- Developed and maintained scripts to automate the collection of key network metrics (e.g., traffic loads, error rates, handover success rates) from network nodes running on Linux.
- Managed site restoration and performed LKF request and installation using ENM and OSS tools, ensuring minimal service disruption and quick recovery.
- Automated daily tasks such as KPI updates, fault tickets, and incident tracking using Python and shell scripting to streamline reporting processes.

RAN Engineer in Jersey Airtel Pvt Ltd - Gurgaon, India

Nov 2020 – Dec 2021

Project: Airtel-Vodafone UK

- Real-time 24x7 monitoring of Airtel-Vodafone (Jersey, UK) RAN node alarms, using tools and network monitoring platforms to ensure network reliability.
- Responsible for fault management of RU, RRU, BB 6620, 6630, 5212, and 5216, working with Linux server logs and tools to identify and troubleshoot performance issues.
- Conducted in-depth traffic trend analysis using PRTG Graphs for Uplink and Downlink, providing insights into network performance and optimizing resources.
- Utilized Python scripting to automate daily outage reports, performance metrics (MoU, ISP usage), and user activation/deactivation reports, improving operational efficiency.
- Managed Incidence tickets and followed up on fault resolution, ensuring timely closure and proper root cause analysis (RCA) in alignment with SLA.
- Coordinated with Field engineer for hardware replacement and faulty components to restore services.

NOC Engineer in Bharti Infratel Ltd - Gurgaon, India

Sep 2019 – Nov 2020

- 24x7 NOC surveillance of network sites, analyzing and resolving faults using network performance data gathered from Linux-based monitoring platforms like NETCOOL and MAXIMO.
- Collaborated with Nokia, Ericsson, ZTE, Huawei, and IBM network teams for OSS integration, fault resolution, and migration, ensuring network performance optimization.
- Monitored network KPIs and managed fault tickets, escalating issues in line with the structured escalation matrix and tracking resolution via OSS tools.
- Developed ad-hoc reports and dashboards for performance monitoring.
- Provided specialized support during emergency situations (e.g., Cyclone) by mapping network parameters on Google Maps, ensuring timely recovery actions.

NOC Engineer in Reliance Jio Infocomm Ltd - Noida, India

Nov 2017 – Aug 2019

- Monitored and troubleshoot 24x7 eNodeB, ISC, and OSC sites for RAN related faults.
- Logged into Cisco routers and managed site link status using tools like PUTTY for command-line interface (CLI) access to network elements, analyzing network uptime and performance.
- Generated and automated progress reports using python scripts, ensuring timely and accurate reporting of network performance and ticket status.

- Managed ticketing systems for fault resolution, ensuring compliance with SLAs and closure of CM and PM tickets.
- Created and published outage SMS, dashboards, and special reports, utilizing Linux-based scripting tools to automate notifications and updates.

Education

- B. Tech. in ECE from Nalanda Institute of Technology, Bhubaneswar, India in 2017 with 7.3 CGPA.
- 12th from Mahatma Gandhi College, Darbhanga in 2013 with 61%.

Declaration

I hereby declare that all the information stated above is true and correct to the best of my knowledge and believe.

Date -
Place –

Chandan Singh